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Specular reflection model for higher multipoles

In this set of notes, we generalize the results of

<https://hackmd.io/@aligho/ryN4ny8WZl> (referred to as **Notes 1** in the following) to higher multipoles (e.g. $l > 1$). In

particular, we derive the results of the paper: [Multipolar response of small metallic spheres: Nonlocal theory](https://journals.aps.org/prb/pdf/10.1103/PhysRevB.35.3722)

(<https://journals.aps.org/prb/pdf/10.1103/PhysRevB.35.3722>) by **Claro** and **Fuchs** (referred to as **Paper 1** in the following). The methodology below is essentially identical to that used in **Notes 1**. In the below, any notation that is undefined may be found either in **Notes 1** or in the paper [Polarizability of a small sphere including nonlocal effects](https://journals.aps.org/prb/pdf/10.1103/PhysRevB.35.3722)

(<https://journals.aps.org/prb/pdf/10.1103/PhysRevB.35.3722>) by **Dasgupta** and **Fuchs**. The only other comment on notation we make is that we use the notation $|\mathbf{k}|$ and k interchangeably in the below. Additionally, we may omit the subscripts (r, E) and (k, E) if it is obvious that the angles we refer to are in the coordinate domain or the Fourier domain.

Section 1: Analogy of Section 1 of Notes 1 (Potential in the Fourier domain)

In this section, we calculate the potential, V_D , in the Fourier domain. This may be done easily by noting that, in analogy with **Section 1** of **Notes 1**, we have:

$$\sigma(\theta) = - \left[\partial_r V_D(\mathbf{r}) \right]_{r=a^+} + \left[\partial_r V_D(\mathbf{r}) \right]_{r=a^-} = P_l(\cos(\theta_{r,E})) \delta,$$

where the only change from **Notes 1** was in the substitution $\cos(\theta_{r,E}) \rightarrow P_l(\cos(\theta_{r,E}))$. By using Laplace's equation for the potential, as in **Notes 1**, we then immediately obtain:

$$V_D(\mathbf{k})|\mathbf{k}|^2 = a^2 \delta \int_{r=a} e^{-i\mathbf{k}\cdot\mathbf{r}} P_l(\cos(\theta_{r,E})) d\Omega$$

This is the generalization of **Equation 11** of the paper Polarizability of a small sphere including nonlocal effects

(<https://journals.aps.org/prb/pdf/10.1103/PhysRevB.35.3722>) by **Dasgupta** and **Fuchs**.

Section 2: Analogy of Section 2 of Notes 1 (Closed form expression for the potential in the Fourier domain)

In this section, we find the closed-form expression for $V_D(\mathbf{k})$ for general values of angular momentum. We write the Legendre polynomials in terms of spherical harmonics as follows (note that for $m = 0$ the associated Legendre polynomials are the same as the Legendre polynomials):

$$P_l(\cos(\theta)) = \left[\frac{4\pi}{2l+1} \right]^{1/2} Y_{l,0}(\theta, \phi)$$

From which we obtain, by expanding the phase factor, $e^{-i\mathbf{k}\cdot\mathbf{r}}$ in partial waves and using the orthogonality of the spherical harmonics:

$$V_D(\mathbf{k}) = \frac{(-i)^l 4\pi a^2 \delta}{|\mathbf{k}|^2} P_l(\cos(\theta_{k,E})) j_l(|\mathbf{k}|a)$$

This is the generalization of **Equation 14** of Polarizability of a small sphere including nonlocal effects (<https://journals.aps.org/prb/pdf/10.1103/PhysRevB.35.3722>) by **Dasgupta** and **Fuchs**.

Section 3: Analogy of Section 3 of Notes 1 (Coordinate space representation of the potential-for $r < a$)

We now go back into coordinate space. Note that, by assumption, the expressions derived by inverse Fourier transforming the expression derived in **Section 2** will only be valid for radial coordinates that lie in the spherical scatterer (i.e. for $r < a$). As in **Notes 1**, we use the established integral identity for the spherical Bessel functions of the first kind from

<https://functions.wolfram.com/Bessel->

[TypeFunctions/SphericalBesselJ/21/02/02/0001/](https://functions.wolfram.com/Bessel-TypeFunctions/SphericalBesselJ/21/02/02/0001/)

(<https://functions.wolfram.com/Bessel-TypeFunctions/SphericalBesselJ/21/02/02/0001/>). (for $r < a$):

$$V_D(\mathbf{r}) = \frac{a^2 \delta(4\pi)}{2\pi} \left[\int_0^\infty j_l(ka) j_l(kr) dk \right] = \frac{a^2 \delta}{2l+1} \frac{r^l}{a^{l+1}} P_l(\cos(\theta))$$

Note that, in deriving this, we also implicitly used the fact that the spherical harmonics, for $m = 0$, are real valued. Analogously, we have, for the total potential:

$$V(\mathbf{r}) = \frac{a^2 \delta(4\pi)}{2\pi} P_l(\cos(\theta)) \left[\int_0^\infty \frac{j_l(ka) j_l(kr)}{\varepsilon(k, \omega)} dk \right]$$

Section 4: Analogy of Section 4 of Notes 1 (Multipolar polarizability in the specular reflection model and derivation of Equation 5 of Paper 1)

At this point, all we need to do is apply Maxwell boundary conditions in order to obtain the multipolar polarizability. Using **Equation 2**, **Equation 3** and **Equation 4** of **Paper 1**, we obtain, for $r = a^+$ (just outside of the sphere):

$$V(\mathbf{r})_{r=a} = C_l \left[\frac{2l+1}{4\pi} \right]^{1/2} \left[a^l - \frac{\alpha_l}{a^{l+1}} \right] P_l(\cos(\theta)) = \frac{E_l^{-1}}{2l+1} (a\delta) P_l(\cos(\theta)),$$

where we introduced the quantity E_l , which is defined in **Equation 6** of **Paper 1**, and the second equality above matches the total potentials just inside and just outside the sphere. Similarly, we may find the displacement field associated with the total potential calculated above and match it with the displacement field associated with $V_D(\mathbf{r} \mid |\mathbf{r}| < a)$:

$$-D(\mathbf{r})_{r=a} = C_l \left[\frac{2l+1}{4\pi} \right]^{1/2} \left[la^{l-1} + (l+1) \frac{\alpha_l}{a^{l+2}} \right] P_l(\cos(\theta)) = \frac{\delta l}{2l+1} P_l(\cos(\theta))$$

At this point, all we need to do is solve for the multipolar polarizability, α_l , which just involves trivial algebraic manipulations. By dividing our second relation with our first relation, we obtain the following:

$$(a^{-1}l)E_l = \frac{la^{l-1} + (l+1)\alpha_l/a^{l+2}}{a^l - \alpha_l/a^{l+1}} \rightarrow E_l = \frac{la^l + (l+1)\alpha_l/a^{l+1}}{la^l - l\alpha_l/a^{l+1}}$$

Thus, we readily obtain:

$$\alpha_l = -\frac{la^l - lE_la^l}{lE_l/a^{l+1} + (l+1)/a^{l+1}} = a^{2l+1} \frac{l(E_l - 1)}{l(E_l + 1) + 1}$$

This is **Equation 5** of **Paper 1**.

Appendix: Notational point

We note that E_l as defined in **Paper 1** is closely related to $F(r)$ as defined in Polarizability of a small sphere including nonlocal effects

(<https://journals.aps.org/prb/pdf/10.1103/PhysRevB.35.3722>) by **Dasgupta** and **Fuchs**. In particular, we have:

$$E_1^{-1} = 3aF(a) \leftrightarrow E_1 = F^{-1}(a)/(3a)$$

Thus, we have, for the dipolar polarizability:

$$\alpha_1 = a^3 \left[\frac{1 - E_1^{-1}}{1 + 2E_1^{-1}} \right] = a^3 \left[\frac{1 - 3aF(a)}{1 + 6aF(a)} \right],$$

which is exactly what we showed in **Notes 1**.